Exercise 1.

(a) Suppose that $f \in C([a,b])$, $f(x) \geq 0$ for all $x \in [a,b]$. Prove that if

$$\int_a^b f(x) = 0,$$

then $f(x) = 0$ for all $x \in [a,b]$.

(b) Suppose that $u \in C([a,b])$ is twice continuously differentiable, $V \in C([a,b])$, $V(x) \geq 0$ for all $x \in [a,b]$ and

$$\begin{aligned} -u''(x) + V(x)u(x) &= 0 \\ u(a) = u(b) &= 0. \end{aligned}$$

Prove that $u(x) = 0$ for all $x \in [a,b]$. *Hint: What's one of the most useful theorems in analysis mentioned in Lecture 22?*

Proof. Starto!

(a) Let $f \in C([a,b])$ and $f(x) \geq 0$ for all $x \in [a,b]$. Assume, for the purpose of proof,

$$\int_a^b f(x) = 0.$$

By the Fundamental Theorem of Calculus, $\exists F \in C([a,b])$ such that

$$F(x) = \int_a^x f(t) \geq 0 \implies F'(x) = f(x) \geq 0$$

and

$$F(a) = \int_a^a f(t) = 0 = \int_a^b f(t) = F(b).$$

By the Derivative Characterization of Monotone Functions, $\forall x, y \in [a,b]$ such that

$$-\infty < x \leq y < \infty \implies -\infty < F(x) \leq F(y) < \infty;$$

however,

$$F(a) = F(b) \implies \exists K \in \mathbb{R}\ \forall x \in [a,b] : F(x) = K \implies F'(x) = 0.$$

Thus, $f(x) = 0$ for all $x \in [a,b]$, as needed.

(b) Let $u \in C^2([a,b])$, $V \in C([a,b])$ and $V(x) \geq 0$ for all $x \in [a,b]$. Assume, for the purpose of proof,

$$\begin{aligned} -u''(x) + V(x)u(x) &= 0 \\ u(a) = u(b) &= 0. \end{aligned}$$

By the Integration by Parts Formula,

$$\int_a^b u^2(x)V(x) = \int_a^b u(x)u''(x) = u(x)u'(x)\Big|_a^b - \int_a^b (u')^2(x) = -\int_a^b (u')^2(x).$$

By the Order-Integral Relation of Continuous Functions,

$$\forall x \in [a,b] : u^2(x)V(x) \wedge (u')^2(x) \geq 0 \implies \int_a^b u^2(x)V(x) \wedge \int_a^b (u')^2(x) \geq 0;$$

hence,

$$\int_a^b (u')^2(x) = 0 \implies \forall x \in [a,b] : (u')^2(x) = 0 \implies u'(x) = 0.$$

Thus, $u(x) = 0$ for all $x \in [a,b]$, as needed.

$\square$

Exercise 2. Compute the derivatives

(a)

$$\frac{d}{dx}\left(\int_{-x}^{x} e^{s^2} ds\right).$$

(b)

$$\frac{d}{dx}\left(\int_{0}^{x^2} \sin(s^2) ds\right).$$

Proof. Let $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$.

(a) Suppose $f(x) = x^2$ and $g(x) = e^x$. Then f and g are continuous on $(-\infty, \infty)$. By the Composition Property of Continuous Functions, $g \circ f$ is continuous on $(-\infty, \infty)$. Then $g \circ f$ is integrable on $[-x, x]$ for all $x > 0$. By the Riemann Integral Theorem,

$$\int_{-x}^{x} e^{s^2} ds = \int_{-x}^{x} (g \circ f)(s) ds \in \mathbb{R}.$$

Since $(g \circ f)(-x) = e^{(-x)^2} = e^{x^2} = (g \circ f)(x)$ for all $x \geq 0$, $g \circ f$ is even on $(-\infty, \infty)$. By the Even Property of Integration,

$$\int_{-x}^{x} e^{s^2} ds = \int_{-x}^{x} (g \circ f)(s) ds = 2\int_{0}^{x} (g \circ f)(s) ds = 2\int_{0}^{x} e^{s^2} ds.$$

Then $\int (g \circ f)$ is differentiable on $\mathbb{R}$. By the Fundamental Theorem of Calculus,

$$\frac{d}{dx}\left(\int_{-x}^{x} e^{s^2} ds\right) = \frac{d}{dx}\left(2\int_{0}^{x} e^{s^2} ds\right) = 2\frac{d}{dx}\left(\int_{0}^{x} e^{s^2} ds\right) = 2e^{x^2}.$$

(b) Suppose $f(x) = x^2$ and $g(x) = \sin x$. Then f and g are continuous on $(-\infty, \infty)$. By the Composition Property of Continuous Functions, $g \circ f$ is continuous on $(-\infty, \infty)$. Then $g \circ f$ is integrable on $[0, x^2]$ for all $x \in \mathbb{R}$. By the Riemann Integral Theorem,

$$\int_{0}^{x^2} \sin(s^2) ds = \int_{0}^{x^2} (g \circ f)(s) ds \in \mathbb{R}.$$

Then $\int (g \circ f)$ is differentiable on $\mathbb{R}$. By the Fundamental Theorem of Calculus,

$$\frac{d}{dx}\left(\int_{0}^{x} \sin(s^2) ds\right) = \frac{d}{dx}\left(\int_{0}^{x} (g \circ f)(s) ds\right) = (g \circ f)(x) = \sin(x^2).$$

Since $f'(x) = 2x$ for all $x \in \mathbb{R}$, f is differentiable on $(-\infty, \infty)$. By the Chain Rule,

$$\frac{d}{dx}\left(\int_{0}^{x^2} \sin(s^2) ds\right) = \frac{d}{df}\left(\int_{0}^{f(x)} \sin(s^2) ds\right) \cdot \frac{df}{dx} = \sin(f^2(x)) \cdot f'(x) = 2x\sin(x^4).$$

$\square$

Exercise 3. Suppose $f : [a, b] \to \mathbb{R}$ is continuous and $\int_a^x f = 0$ for all rational $x \in [a, b]$. Show that $f(x) = 0$ for all $x \in [a, b]$.

Proof. Let $f \in C([a, b], \mathbb{R})$. Suppose, for the remainder of the proof, $\int_a^x f = 0$ for all $x \in [a, b] \cap \mathbb{Q}$. By the Fundamental Theorem of Calculus, $\exists F : [a, b] \to \mathbb{R}$ such that

$$F(x) = \int_a^x f(t) \implies F'(x) = f(x).$$

Then F is differentiable and continuous everywhere. We want to show that $F(x) = 0$ for all $x \in [a, b]$. Assume, for the purpose of contradiction, $F(y) \neq 0$ for some $y \in [a, b] \setminus \mathbb{Q}$. Choose $\epsilon_0 = |F(y)|$ and $\delta > 0$. By the Density of $\mathbb{Q}$, $\exists x \in [a, b] \cap \mathbb{Q}$ such that $|x - y| < \delta$ and

$$|F(x) - F(y)| = |0 - F(y)| = |F(y)| \geq \epsilon_0.$$

Then F is discontinuous at $y \in [a, b]$, which is a contradiction. We have shown that $F(x) = 0$ for all $x \in [a, b]$. By the Derivative Characterization of Monotone Functions,

$$F(x) = 0 \implies F'(x) = 0.$$

Thus, $f(x) = 0$ for all $x \in [a, b]$, as needed. $\square$

Exercise 4.

(a) Find the pointwise limit of $e^{x/n}/n$ for all $x \in \mathbb{R}$.

(b) Is the limit uniform on $\mathbb{R}$?

(c) Is the limit uniform on $[0, 1]$?

Proof. Let $f_n : \mathbb{R} \to \mathbb{R}$ and $f : \mathbb{R} \to \mathbb{R}$.

(a) Suppose, for the remainder of proof, $f_n(x) = e^{x/n}/n$ and $f(x) = 0$. By the Product Theorem of Convergent Sequences,

$$\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \frac{e^{x/n}}{n} = \lim_{n\to\infty} e^{x/n} \cdot \lim_{n\to\infty} \frac{1}{n} = 1 \cdot 0 = f(x).$$

(b) Choose $\epsilon_0 = 1$ and $M \in \mathbb{N}$. By the Surjective Property of Exponential Functions, $\exists x_0 \in \mathbb{R}$ such that $x_0/M = \ln M$ and

$$|f_M(x_0) - f(x_0)| = \left| \frac{e^{x_0/M}}{M} - 0 \right| = \frac{e^{x_0/M}}{M} = \frac{e^{\ln M}}{M} = 1 \geq \epsilon_0.$$

(c) Choose $\epsilon > 0$ and $x \in [0, 1]$. By the Archimedean Property of Real Numbers, $\exists M \in \mathbb{N}$ such that $M\epsilon > e$ and

$$\forall n \geq M : |f_n(x) - f(x)| = \left| \frac{e^{x/n}}{n} - 0 \right| = \frac{e^{x/n}}{n} < \frac{e}{M} < \epsilon.$$

$\square$

Exercise 5. Suppose $\{f_n\}$ and $\{g_n\}$ defined on some set S converge to f and g respectively uniformly on S. Show that $\{f_n + g_n\}$ converges uniformly to $f + g$ on S.

Proof. Let $f_n : S \to \mathbb{R}$ and $g_n : S \to \mathbb{R}$. Assume $f : S \to \mathbb{R}$ and $g : S \to \mathbb{R}$ such that $f_n \to f$ and $g_n \to g$ uniformly. We want to show that $f_n + g_n : S \to \mathbb{R}$ converges uniformly to $f + g : S \to \mathbb{R}$. Fix $\epsilon > 0$. By the Uniform Convergence of f_n and g_n, $\exists M_f, M_g \in \mathbb{N}\ \forall x \in S$ such that

$$\forall n \geq M_f : |f_n(x) - f(x)| < \frac{\epsilon}{2} \wedge \forall n \geq M_g : |g_n(x) - g(x)| < \frac{\epsilon}{2}.$$

Choose $M = M_f + M_g$. By the Triangle Inequality of $\mathbb{R}$,

$$\begin{aligned} \forall n \geq M\ \forall x \in S : |(f_n + g_n)(x) - (f + g)(x)| &= |f_n(x) - f(x) + g_n(x) - g(x)| \\ &\leq |f_n(x) - f(x)| + |g_n(x) - g(x)| < \epsilon. \end{aligned}$$

Thus, $f_n + g_n \to f + g$ uniformly, as needed. $\square$